Air Pollution – A Hidden Menace

A. Lead in :

Has it ever happened to you that when you come back home from outside, you have a running nose or you keep on coughing ? Have rain drops ever tasted sour ? When you are on public roads, you inhale a lot of polluted air and you feel uneasy. Many factors contribute to this air getting polluted. Air pollution is a hidden menace and poses the greatest threat to mankind in the future. Let us read the following piece and think of ways in which we can ensure that we breathe clean and pure air.

B. The Text :

No one can forget one of the most tragic **industrial** accidents that occurred at Bhopal on 3 December, 1984. Deadly gas from a chemical plant operated by **Union Carbide** escaped into the atmosphere, killing over 4000 local residents and rendering blind and **crippling** a large section of the city's surviving population. Not only Bhopal but now every city, every town, every corner of the earth is facing such a crucial problem. Every day, every moment we breathe polluted air and may become a **victim** of air pollution.

A man can live without food for a month, without water for two or three days, but he cannot live without breathing even for a minute. It is estimated that an average adult exchanges 15 kg of air a day, in comparison to about 1.5 kg of the food consumed and 2.5 kg of water **intake**. It is obvious that the quantum of pollutants that enter our body through respiration would be **manifold** in comparison to those taken in through polluted water or **contaminated** food.

Air is a mixture of gases comprising 78 percent nitrogen, 21 percent oxygen and a little less than 1 percent **argon**, together with 0.03 percent carbon dioxide. These elements make up 99.9 percent of dry air. As long as this composition is maintained, the air is pure. If this composition is altered, i.e. the oxygen level gets reduced or irritating gases enter the atmosphere, then the air is said to be polluted and inhalation of this polluted air can lead to respiratory disorders.

Our air is being poisoned with the by-products of an expanding technological society. Air pollution is nothing new, but what is new is the scope and **severity** of air pollution.

In recent times, quite a large number of industries can be seen in urban areas as well as in rural pockets. Most of these industries **spew** dense smoke from their chimneys. What is this smoke made of and how is it produced ? Industries require steam and to produce it various fuels such as coal, coke, furnace oil are burnt. During burning, along with heat, smoke is also produced. Where does this smoke go ? Apparently, it disappears in a short time but in reality it never does so. Instead, it mingles with the atmospheric air and pollutes it. We respire this polluted air containing **obnoxiou**s gases, ash and dust particles. Without our knowledge, our lungs slowly become garbage dumps for these pollutants.

Thermal power stations are rated the first among the industries that discharge high amounts of smoke and ash. Other significant industries contributing to air pollution are cement, steel and ore processing industries. Some of the chemical industries also release toxic fumes into the air, along with smoke.

The automobile **exhausts** are in no way less dangerous than the industrial smoke. It is reported that automobiles in Greater Kolkata alone spew about 1500 tonnes of pollutants into the atmosphere every day. It is stated that a person living in Kolkata, whether he is a smoker or not, is forced to inhale toxic substances equivalent to smoking two packets of cigarettes a day. The levels of pollution in cities like Delhi, Mumbai and Chennai are equally alarming. To meet the demands of an exploding population, the number of buses **plying** on the roads are being increased. Equally a greater number of lorries and other goods carriers are on the move. Along with heavy vehicles, use of cars, jeeps and two-wheelers such as bikes, scooters and mopeds have increased dramatically – all contributing to significant levels of air pollution. Automobiles are responsible for 60 percent of air pollution in various parts of the world as they release maximum carbon monoxide into the atmosphere. The **menace** of air pollution attributed to the automobile exhausts has now reached the peak level and if this trend continues, we may have to wear nasal filters on our nose in future.

The damage caused by pollution is enormous. In money alone it represents a loss of billions of dollars each year. Many flower and vegetable crops suffer ill effects from car exhaust gases. Trees have been killed by pollution from power plants. Cattle have been poisoned by the fumes from **smelters** and recover aluminum from ore. Air pollution causes rubber tyres on automobiles to crack and become **porous**. Fine buildings become shabby, their walls blackened with soot that has settled on them. Building surfaces may actually deteriorate because of air pollution.

But the high cost of air pollution is most strikingly illustrated in its damaging effects on the human body. Air pollution causes eye irritations, scratchy throats, and respiratory illnesses. It also contributes to a number of serious diseases. In both the United States and Europe, periods of high levels of air pollution were linked to an increased number of deaths.

Much direct harm is done by air pollution. Scientists are alarmed because the amounts of gases such as carbon dioxide, methane, and nitrous oxide in our atmosphere are increasing. These gases tend to trap the radiation that reaches the earth from the sun and as a consequence of which the atmosphere could become warmer. This process would eventually lead to global warming.

Scientists have been concerned, too, about the widespread use of a substance that may destroy the atmospheric layer that protects us from harmful kinds of solar energy. This substance belongs to a group of chemicals and **chlorofluorocarbons**. It is used as a refrigerant and a cleaner and was once widely used in spray cans.

Another concern is acid rain. This is rain or other **precipitation** that contains oxides of sulphur and nitrogen, along with other chemicals. Acid rain causes damage in lakes and rivers. It poisons the plants and animals that live in the water. It may also affect crops and other plants, stone buildings and monuments and drinking water.

Acid rain affects everything it falls on. The water in rivers and lakes turns acidic. For instance, in Sweden, 4000 lakes have been so severely affected that no fish has survived. It also changes the soil's nutrient content. It washes or **leaches away** nutrients like potassium, calcium and magnesium from the upper layer that help trees grow. Acid rain kills large stretches of forests, leaving behind leafless skeletons of trees.

When forests begin to die, the animals and birds in those forests follow. Among the growing list of species threatened by acid rain are the Pied Flycatcher and Apollo Butterfly in Sweden. The Dipper fish has vanished from the river of Central Wales, and the Brown Trout from Norwegian lakes. The list goes on.

What about our health ? Acid rain irritates the sensitive tissues of our eyes and lungs, particularly in children. It can also cause skin **lesions.**

Living beings apart, even buildings are not spared. In Poland, the beautiful old buildings of Krakow are slowly being destroyed by acidic **smog**. In Athens, a city which is highly polluted, acid rain is eating into the marble of its world-famous monuments. Experts say that more damage has been done in the past 25 years than in the previous 2000 !

There are three basic approaches to control air pollution – Preventive measures, such as changing the raw materials used in industry or the **ingredients** of fuel; **dispersal measures** such as raising the heights of **smokestacks**; and collection measures, such as designing equipment to **trap** pollutants before they escape into the atmosphere.

Nearly, all the highly industrialized countries of the world have some type of **legislation** to prevent and control air pollution. One difficulty is that pollutants may be carried by the wind from one country to another, often for distances of thousands of miles. The death of lakes in eastern Canada has been caused by acid rain that originated in the United States. Acids produced in Britain and France have caused damage in Sweden.

There have been many initiatives in different countries for making law, setting standards and norms to check air pollution and ensure quality air. Air quality programmes have brought improvements in many areas. For example, burning low-sulphur coal and oil in factories and power plants has lowered pollution in many cities. To meet standards, automobile engines have been re-designed and new cars have been equipped with devices such as the catalytic converter which changes pollutants into harmless substances. Because of these new devices, air pollution from car exhaust has also been reduced.

It is not easy to bring about the new developments needed to control air pollution. Many people – physicians, engineers, **meteorologists**, botanists, and others are involved in research, seeking new ways. Vast sums of money will have to be spent in the future to clean the air and to keep it clean. Often pollution control means higher prices – to cover the cost of control devices in **emission systems** of new cars, for example. But to most people, the cost is justified. Perhaps the day will come when people everywhere can breathe pure air in cities where the sunlight is no longer blocked by an umbrella of pollution.

C. Notes and glossary :

industrial	:	relating to industry
Union Carbide	:	name of the industry in Bhopal
crippling	:	damaging
victim	:	someone who suffers as a result of something
intake	:	consumption

manifold	:	of many different kinds
contaminated	:	impure
argon	:	chemically inactive gas
severity	:	seriousness
spew	:	throw out
obnoxious	:	unpleasant
exhausts	:	the gas or steam out of the engine of a car etc.
plying	:	running
menace	:	threat
smelters	:	furnaces
porous	:	having small holes
chlorofluorocarbons	:	chemicals used for cooling in refrigerators
precipitation	:	fall of rain, snow or hail
leaches away	:	washes away
lesions	:	wounds or injuries
smog	:	a mixture of smoke and fog
ingredients	:	things used to make something
dispersal measures	:	ways of scattering things
smokestacks	:	tall chimneys that carry smoke away from factories
trap	:	retain
legislation	:	a body of laws
meteorologists	:	persons who study weather conditions
emission system	:	a system of sending out smoke

D. Let's understand the text :

1. What accident took place at Bhopal in 1984 ?
2. Why is it called an industrial accident ?
3. What were the tragic consequences of it ?
4. How is air important for man ?
5. What is the major source of contamination of the human body ?
6. What is the composition of air ?
7. When is air said to be polluted ?
8. What fuels do the industries use ?
9. How does the released smoke affect man ?
10. Why do thermal power stations cause more pollution ?
11. What are the other industries equally harmful for us ?
12. Why are the automobiles increasing on the road ?
13. How do the automobiles contribute to air pollution ?
14. What are the effects of air pollution on vegetable crops and trees ?
15. How are buildings affected by polluted air ?
16. What health problems are caused by polluted air ?
17. How is air pollution responsible for increasing the temperature ?
18. What harm can refrigerant do ?
19. What is acid rain ?
20. How is water affected by acid rain ?
21. What is the impact of acid rain on soil ?
22. What are the ways to control air pollution ?
23. How have different countries tried to check it ?
24. How have the air quality programmes brought us benefits ?
25. Why do we still need to find out better ways to control air pollution ?

E. Let's go beyond the text :

(i) 1. Why does the oxygen level in the atmosphere get reduced ? What could be its consequences ?

2. How do the chemical industries cause dangerous air pollution ? What is its far reaching consequences ?

3. How is acid rain caused ?

4. How does polluted air travel from one country to another ?

5. What steps can be taken to reduce the pollution caused by automobiles`? Is air pollution a global problem ?

(ii) Given below is a table. Read the text and complete the table.

	Caused by	**Impact on**	**Nature of Damage**
	industries		
			smoke and ash in air
Air Pollution	automobile		
		soil	
			destroying atmosphere layers

F. Let's do some activities :

1. Let's think together.

The text you have read mentions some steps to reduce air pollution. Now work in groups to suggest more steps for dealing with the problem. You can refer to newspapers, journals and discuss among yourselves to get more ideas (The teacher divides the class into four or five groups for the purpose)

2. Let's speak and listen.

Each group presents its ideas in the class regarding the steps to control air pollution. Other groups listen and react to the ideas.

After all the groups have presented ideas, commonly agreed aspects should be finalized (The teacher acts as the observer and coordinator.)

3. Let's write :

Write a letter to the Editor of a newspaper / the Minister, Department of Environment explaining the importance of clean air and suggesting ways to keep it clean and pure.

4. Let's debate:

Some speak 'for' the motion and some 'against'. The teacher acts as Chairperson.

i) Use of private vehicles should be banned to control air pollution.

ii) Industries being the major source of pollution need to be disestablished.

G. Let's enrich our vocabulary :

i) There are some words / phrases in the text used to convey the harmful effects of air pollution on mankind.

A few examples are given : deadly, irritating gases

Pick out the other such words / expressions from the text and write in your copy.

ii) (a) The word 'respiration' is the noun form of the verb 'respire' : Now with the help of a dictionary find out verb / noun forms of the following.

Noun	**Verb**	**Verb**	**Noun**
Resident		Consume	
Pollution		Alter	
Composition		Believe	

Inhalation	Reduce
Emission	Produce
Comparison	Operate
Knowledge	Expect
Recovery	Represent
Destruction	Radiate
Contamination	Illustrate
Precipitation	Justify
Deterioration	Prevent

(b) Arrange the words provided under **noun** and **verb** in the order in which they should come in a dictionary.

iii) Read the sentences below :

………. buildings of Krakow are slowly being destroyed by acidic smog.

Mark the word underlined. Two words, i.e. 'smoke and fog' (smoke + fog) have formed the word 'smog'. Such process of word formation is known as **blending**. Many new words are being made in this process and are increasingly in use. Given below is an exercise. Complete it.

Foreign + exchange bank = ……………… bank

…………….. + ……………… policy = exim policy

Slim + tender = …………………

…………….. + ……………… = telecast

iv) Given below are some expressions. Use a single word for each expression. Go to the text to find the words.

a great threat

getting worse day by day

things used to make something

a body of laws

one who studies weather conditions.

H. Let's learn language :

a) Mark the following sentences used in the text :

i) ……….. irritating gases enter the atmosphere.

ii) Every day, every moment we breathe polluted air to become a victim of air pollution.

The words underlined above are known as 'participle adjectives'.

Here 'irritating' is the present participle adjective and 'polluted' is the past participle adjective in the above sentences.

Find out in the text how many such participles are there and make a list.

Present Participle __________, __________, __________, __________,

Past Participle __________, __________, __________, __________,

Complete the following sentences with appropriate participles of the verbs given in brackets.

1. The beggar is wearing a ……………….. shirt. (tear)

2. My father bought me a ……………….. suit. (swim)

3. My friend wanted a ……………….. instrument .(clean)

4. I never like to take a ……………….. egg. (boil)

5. Air pollution causes ……………….. problem. (breathe)

6. People get deceased in a ……………….. atmosphere. (pollute)

7. Air pollution has ……………….. effect on buildings. (damage)

8. Severity of air pollution is more found in ……………….. countries. (develop)

b) Punctuate the following text :

In the United States control of air pollution is chiefly the responsibility of the state and local governments all the states have air quality management programmes which are patterned after federal laws the basic federal law dealing with air pollution is the clean air act of 1970 amended in 1990 under this law the federal environment protection agency sets standards for air quality what are the standards.

I. Let's prepare a Project :

You live in a locality. You might have experienced some sort of pollution in your locality. It might be pollution of air or water or soil or could be noise pollution. Survey your area with a focus on the **nature of pollution**, **its ill effects, causes of pollution** and **measures to control** the same. Analyse and interpret the data/ information collected. Write all these in a project format.

(It could be an individual or group project.)

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